INCH-POUND

MIL-B-49436/7A(ER)
10 December 1993
SUPERSEDING
MIL-B-49436/7(ER)
3 August 1983

# MILITARY SPECIFICATION SHEET

## BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM BB-507/U

This specification sheet is approved for use by the Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49436.

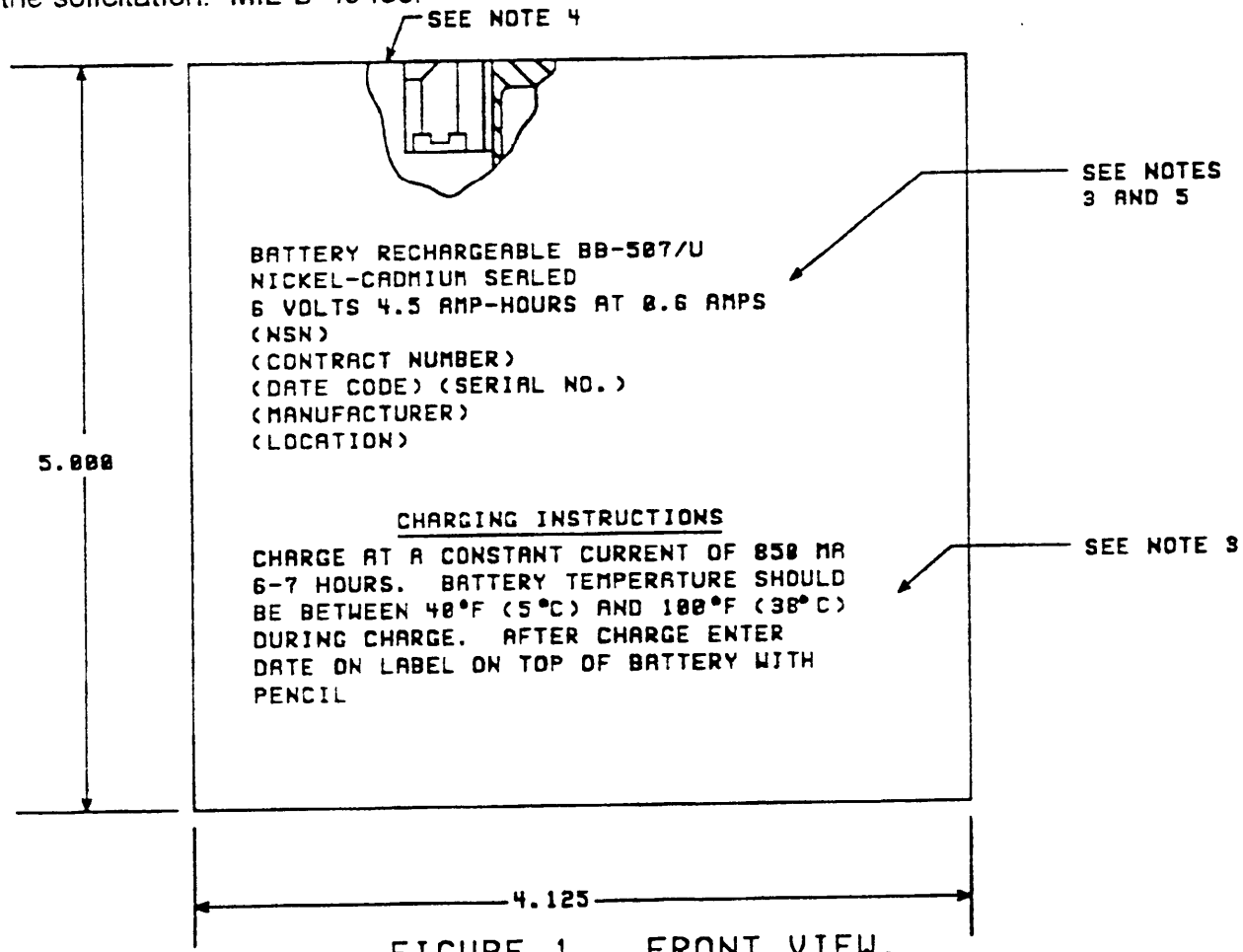



FIGURE 1. FRONT VIEW.



DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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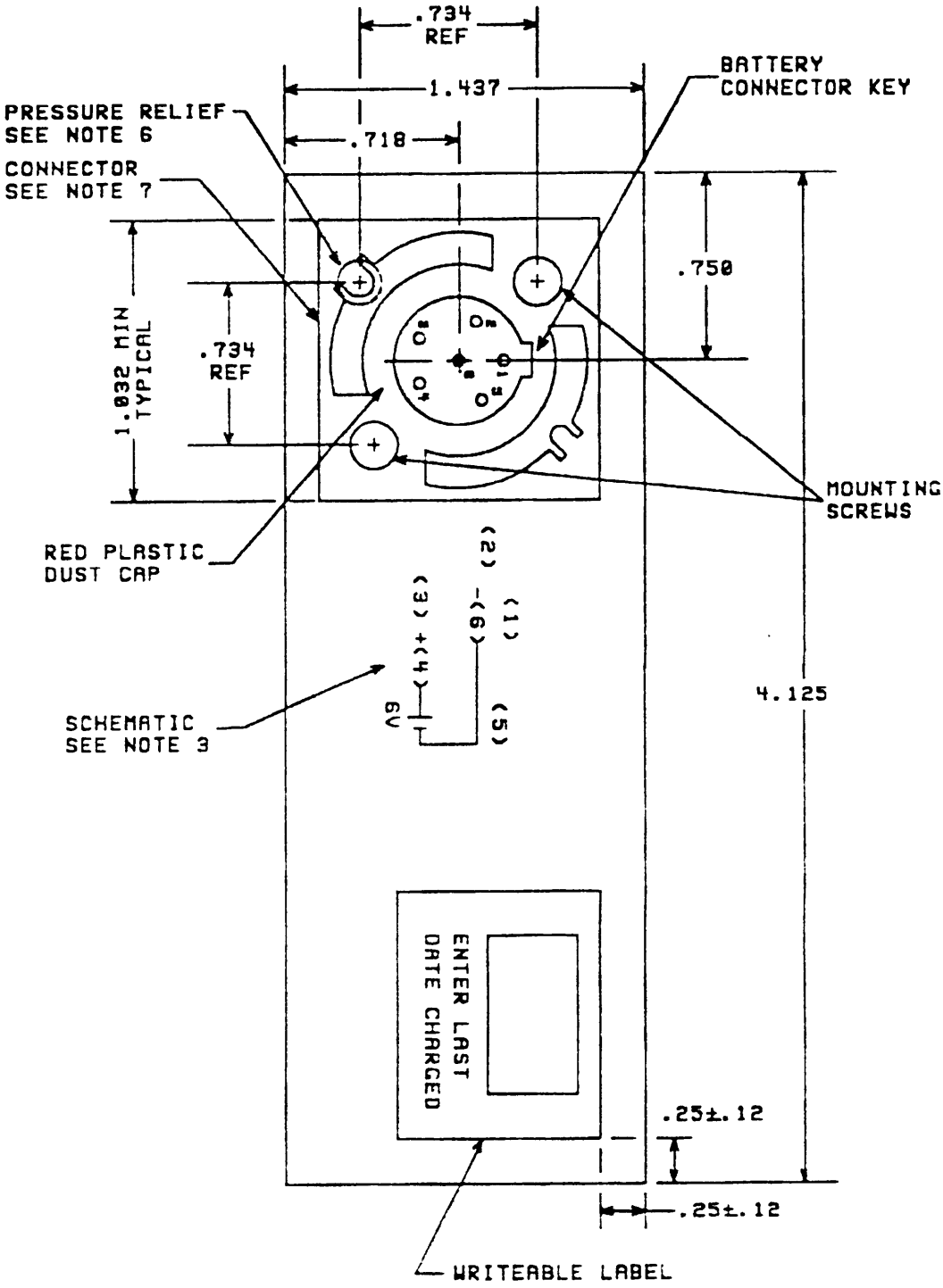



FIGURE 2. TOP VIEW



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## DRAWING NOTES

1. All dimensions are in inches
2. Tolerances on decimals: ± .031 inches.
3. Stencil (or equivalent) schematic, identification and charging instructions as shown. Color marking white.
4. Face of connector shall be flush to .031 inch below top of battery.
5. Manufacturer must fill in applicable information in parenthesis. Location is optional.
6. Pressure relief valve shall be located under connector. It shall release gas before case damage is incurred. It shall not permit the entry of liquid into the battery case.
7. The main connector key shall be oriented as shown in Figure 2. The two connector mounting screws shall be corrosion resistant steel, class 303 per QQ-S-763. Connectors, which are functionally equivalent to the SC-C-179495 and will not cause a reduction in battery performance or life, are acceptable.

REQUIREMENTS:

Rated capacity (C/5): 4.5 AH.

Voltage (Nominal): 6.0 Volts.

Weight (Maximum) 0.95 Kg (2.1 lbs).

Connector: Floating type per SC-C-179495 or equivalent.

Number and type of cells: 5 sealed "D" size cells or equal in series.

Charge rate for testing
and on battery (C/5 for 6-7 hrs) : 0.85 amps.
or when permitted, per 4.8.1.2

| Discharge Rates | 5 Second Voltage(V) | Final Voltage(V) | Current (Amps) |
|---|---|---|---|
| C/5 | - | 5.0 | 0.85 |
| $C_5$ | - | 5.0 | 4.0 |
| 3 $C_5$ | 5.5 | 5.0 | 12.0 |
| 25A | - | 5.0 | 25.0 |



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| Test | Discharge Rate | Capacity Requirement (AH) |
|---|---|---|
| Full Capacity Discharge | $C_5$ | 4.0 |
| Cycle Life, 224 cycles | $C_5$ | 3.2 |
| Overcharge, 24 hours | $C_5$ | 4.0 |
| High Rate Discharge | 3 $C_5$ | 3.0 |
| Low Temperature Discharge at -30°C (-22°F) | $C_5$ | 2.0 |
| Retention of Charge 7 days at 50°C (122°F) | $C_5$ | 2.4 |
| Pulse Discharge 25 amps, 5 seconds on, 25 seconds off, continuously to 5.0 volts | 25A | 2.5 |
| Vibration | C/5 | 4.5 |

NOTES - 1. Unless otherwise specified, all values are minimum.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL:

CUSTODIAN:
ARMY-ER

REVIEWER:
ARMY - CR, AV, GL

PREPARING ACTIVITY:
ARMY-ER

(PROJECT 6140-A787)



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